

Chase J. Sakitis, PhD

Postdoctoral Academic Scholar

Children's Mercy Kansas City
Population Health & Outcomes Research
Fridley Lab

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RESEARCH SUMMARY

My **mission** is to develop rigorous, uncertainty-aware statistical methods that enable accurate, reproducible, and clinically meaningful biomedical discovery. My **research agenda** lies at the intersection of Bayesian statistics, biomedical imaging, spatial omics, and computational biology, where I develop methodologies that strengthen the analytical pipeline from image acquisition and enhancement to statistical modeling and biological interpretation. By integrating advances in Bayesian modeling, computational imaging, and image-driven inference, I seek to improve the reliability and interpretability of complex biomedical data while advancing our understanding of cancer biology and the tumor microenvironment. My **vision** is a future in which multimodal imaging data, interpretable computational models, and principled uncertainty quantification work together to transform complex biomedical information into actionable scientific knowledge and accelerate precision medicine.

EDUCATION

2021 – 2024	Ph.D. in Computational Sciences , <i>Marquette University</i> , Milwaukee, WI
2020 – 2020	M.S. in Applied Statistics , <i>Marquette University</i> , Milwaukee, WI
2014 – 2018	B.A. in Mathematics , <i>Bloomsburg University</i> , Bloomsburg, PA

AWARDS AND HONORS

2024	Travel Award: Marquette University, Graduate School, <i>American Statistical Association Statistical Methods in Imaging 2024</i>
2023	Travel Award: Marquette University, Graduate School, <i>American Statistical Association Joint Statistical Meeting 2023</i>
2022	Travel Support , ISBA and University of California Santa Cruz, <i>O'Bayes 2022: Objective Bayes Methodology Conference</i>
2021	S.J. Merrill Award: Best Teaching Assistant , Marquette University, Klinger College of Arts & Sciences

RESEARCH EXPERIENCE

07/2025 – Present	TL1 Postdoctoral Scholar , Children's Mercy Kansas City/University of Kansas Medical Center, Fairway, KS - Developing Bayesian immune subtyping models using spatial proteomic imaging data to identify prognostic immune phenotypes associated with survival outcomes in epithelial ovarian cancer. - Applied a functional data analysis framework to model spatial clustering trajectories from spatial proteomic images, integrating
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- abundance and spatial organization of T cell populations to evaluate prognostic associations in epithelial ovarian cancer.
- 08/2024 – 06/2025 **Post Doctoral Research Scholar**, Children's Mercy Kansas City, Population Health & Outcomes Research, Fridley Lab, Kansas City, MO
- Developed Bayesian hierarchical models and accompanying software to estimate associations between patient-level predictors, exposures, and immune cell infiltration patterns in spatial single-cell proteomic imaging data.
- 05/2021 – 05/2024 **Graduate Research Assistant / Summer Research Fellow**, Marquette University, Department of Mathematical and Statistical Sciences, Milwaukee, WI
- Developed Bayesian methods for parallel magnetic resonance image reconstruction, including the Bayesian approach to Sensitivity Encoding (BSENSE), Bayesian Generalized Autocalibrating Partial Parallel Acquisition (BGRAPPA), and Bayesian Merged Utilization of GRAPPA and SENSE (BMUGS).
 - Conducted simulation studies and applied Bayesian image reconstruction methods to experimental fMRI data, evaluating performance relative to traditional SENSE and GRAPPA approaches.
 - Developed statistical software and computational workflows for high-dimensional neuroimaging analyses and Bayesian inference.
- 08/2017 – 05/2018 **Independent Research Study**, Bloomsburg University, Department of Mathematics, Computer Science, and Digital Forensics, Bloomsburg, PA
- Applied the Log-Lindley Normal distribution to statistical modeling and risk assessment in toxicology.

EXTERNAL AND INTERNAL FUNDING

Active

Account: TL1TR002368
 Name of PI: **Sakitis, Chase J.**
 Dates: 07/01/2025 – 06/30/2027
 Source: NIH/NCATS
 Title: *Bayesian models for the analysis of spatial proteomic data with application to ovarian cancer*
 % Effort: 100%
 Role in Study: **Principal Investigator**
 Total Award: ~\$162,000

Submitted

Account: 1K99CA321407-01
 Name of PI: **Sakitis, Chase J.**
 Dates: 12/01/2026 – 11/30/2030
 Source: NIH/NCI
 Title: *Bayesian Imaging Models for Enhanced Characterization of the Tumor Microenvironment*
 % Effort: 100%
 Role in Study: **Principal Investigator**

Total Award: ~\$881,596
Score: Not Discussed

RESEARCH INTERESTS

Bayesian Statistical Methodology	Biomedical Imaging and Image-Derived Data Analysis
Statistical Methods for Spatial Omics	Cancer Biostatistics
Survival Analysis	Statistical Machine Learning
Tumor Microenvironment Analysis	Immune Cell Profiling and Subtyping

PEER-REVIEWED PUBLICATIONS

Articles

1. **Sakitis CJ**, Soupir AC, Reid BM, Lawson AB, Schildkraut JM, Peres LC, Fridley BL. *Integrative Bayesian Clustering Models for Immune Subtyping of Spatial Proteomic Imaging Data*. Target Journal: PLOS Computational Biology. **Near Submission**
2. **Sakitis CJ**, Liao D, Reid BM, Townsend MK, Schildkraut JM, Lawson AB, Tworoger SS, Terry KL, Peres LC, Wrobel J, Soupir AC, Fridley BL. *Functional Data Analysis of Spatial Clustering Identifies Prognostic T Cell Patterns in Ovarian Cancer*. PLOS Computational Biology. **Revisions Invited**
3. Fang WB, O'Donnell L, Rekowski M, Clark ZS, Lorenzi P, Tan L, Fridley BL, **Sakitis CJ**, Cheng N. *Network models generated from metabolomics and proteomics profiling reveal the TCA cycle, riboflavin metabolism, glyoxylate and dicarboxylate metabolism, and ribosomal assembly as central pathways that distinguish pre-invasive from invasive breast carcinomas*. Cancers. **Revisions Invited**
4. **Sakitis CJ**, Laborde J, Wrobel J, Soupir AC, Colin-Leitzinger CM, Bitler BG, Townsend MK, Lawson AB, Schildkraut JM, Tworoger SS, Terry KL, Peres LC, Fridley BL. [Bayesian multi cell type models for the analysis of complex immune cell populations with application to ovarian cancer](#). Briefings in Bioinformatics, Volume 27, Issue 1, January 2026, bbag053, doi: 10.1093/bib/bbag053
5. **Sakitis CJ**, Rowe DB. [Bayesian merged utilization of GRAPPA and SENSE \(BMUGS\) for in-plane accelerated reconstruction increases fMRI detection power](#). Magnetic Resonance Imaging, 2025. doi: 10.1016/j.mri.2024.110252
6. **Sakitis CJ**, Rowe DB. [A Bayesian Approach to GRAPPA Parallel FMRI Image Reconstruction Increases SNR and Power of Task Detection](#). Annals of Applied Statistics, 19(2):1473-1493, 2025
7. **Sakitis CJ**, Brown DA, Rowe DB. [A Bayesian complex-valued latent variable model applied to functional magnetic resonance imaging](#). Journal of the Royal Statistical Society Series C: Applied Statistics, 74(1): 100-125, 2025.

Reviews

1. Sabitov K, **Sakitis CJ**, Lesniak N, Soupir AC, Baladandayuthapani V, Fridley BL. *Processing of multiplex spatial protein images*. Target Journal: Nature Reviews Methods Primers. **In Progress**
2. **Sakitis CJ**, Rowe DB. *Bayesian Modeling of Complex Latent Structures in Regression Analysis*. WIREs Computational Statistics. **Under Review**

SOFTWARE AND OPEN-SOURCE RESOURCES

BTIME (CRAN, GitHub)

- R package for Bayesian modeling of the tumor immune microenvironment.

CRAN - <https://cran.r-project.org/web/packages/BTIME/index.html>

GitHub - <https://github.com/FridleyLab/BTIME>

Ovarian_Cancer_FDA (GitHub)

- Public repository containing reproducible code, documentation, and analytic workflows supporting *Functional Data Analysis of Spatial Clustering Identifies Prognostic T Cell Patterns in Ovarian Cancer*.

GitHub - https://github.com/FridleyLab/Ovarian_Cancer_FDA

CONFERENCE PRESENTATIONS AND ABSTRACTS

Oral Presentations

Sakitis CJ, Soupir AC, Reid BR, Lawson AB, Schildkraut JM, Peres LC, Fridley BL. *A Bayesian Model for Tumor Immune Subtyping using Spatial Proteomic Imaging Data*. Joint Statistical Meeting, Section on Statistics in Genomics and Genetics, Boston, MA, 2026.

Sakitis CJ, Soupir AC, Reid BM, Lawson AB, Schildkraut JM, Peres LC, Fridley BL. *Tumor Immune Subtyping Using a Bayesian Model with Application to Ovarian Cancer*. Midsouth Computational Biology & Bioinformatics Society, Tampa, FL, 2026. **[Invited Session]**

Sakitis CJ, Soupir AC, Townsend MK, Laborde J, Johnson CE, Lawson AB, Schildkraut JM, Tworoger SS, Terry KL, Peres LC, Fridley BL. *Assessment of different Covariance Structures to model the Relationship Between Different Immune Cell Populations within a Bayesian Hierarchical Model for Spatial Protein Imaging Data*. Joint Statistical Meeting, Biometrics Section, Nashville, TN, 2025.

Sakitis CJ, Laborde J, Soupir AC, Townsend MK, Lawson AB, Schildkraut JM, Tworoger SS, Terry KL, Peres LC, Fridley BL. *Bayesian Hierarchical Model for Analyzing Spatial Single Cell Protein Imaging Data*. International Biometric Society Eastern North American Region Annual Meeting, New Orleans, LA, 2025.

Sakitis CJ, Rowe DB. *A Bayesian Approach to Fused GRAPPA and SENSE MR Image Reconstruction*. Proc. Joint Statistical Meeting, Section in Imaging, Portland, OR, 2024.

Sakitis CJ, Rowe DB. *Formal Bayesian Approach to GRAPPA Image Reconstruction*. Proc. Joint Statistical Meeting, Section in Imaging, Toronto, Ontario, Canada, 2023.

Sakitis CJ, Brown DA, Rowe DB. *A Full Bayesian Approach to SENSE Image Reconstruction Increases Brain Tissue Contrast and Reduces Noise for More Accurate Statistical Analysis*. Proc. Joint Statistical Meeting, Section in Imaging, Washington, DC, 2022.

Sakitis CJ, Brown DA, Rowe DB. *A Formal Bayesian Approach to SENSE Image Reconstruction*. Proc. Joint Statistical Meeting, Section in Imaging, 25: 1332-1358, Seattle WA (Virtual), 2021.

Posters

Sakitis CJ, Laborde J, Wrobel J, Soupir AC, Colin-Leitzinger CM, Bitler BG, Townsend MK, Lawson AB, Schildkraut JM, Tworoger SS, Terry KL, Peres LC, Fridley BL. *Multi-cell type model for analyzing spatial single-cell protein imaging data with application to ovarian cancer*. American Association for Cancer Research, San Diego CA, 2026.

Sakitis CJ, Rowe DB. *Formal Bayesian Approach to a Fused GRAPPA and SENSE Parallel Imaging Technique Augmenting Task Detection Power*. American Statistical Association Statistical Methods in Imaging, Indianapolis IN, 2024.

Sakitis CJ, Rowe DB. *A Full Bayesian Approach to GRAPPA Reduces Noise in fMRI Image Reconstruction*. American Statistical Association Statistical Methods in Imaging, Minneapolis MN, 2023.

Sakitis CJ, Brown DA, Rowe DB. *A Formal Bayesian Approach to SENSE Image Reconstruction Leads to More Statistically Significant Task Activation in fMRI*. International Society for Bayesian Analysis, Section in Objective Bayes, Santa Cruz CA, 2022.

Sakitis CJ, Brown DA, Rowe DB. *A Full Bayesian Approach to SENSE Image Reconstruction Increases Brain Tissue Contrast and Reduces Noise Leading to More Statistically Significant Task Activation*. American Statistical Association Statistical Methods in Imaging, Nashville TN, 2022.

ADDITIONAL PRESENTATIONS

Invited Talks

Bayesian Multi-Scale Modeling of the Tumor Microenvironment: Linking Cell-Type Interactions, Patient Characteristics, and Survival in Epithelial Ovarian Cancer. Department of Biostatistics and Data Science, University of Kansas Medical Center, Seminar Series, Kansas City, KS, June 2026.

Bayesian Statistical Modeling for Analyzing Single-Cell Protein Imaging Data. Research Month, Children's Mercy, Post Doc Talk, Kansas City MO, May 2025.

Institutional Presentations

Bayesian Complex Valued Latent Regression. Department of Mathematical and Statistical Sciences, Marquette University, Guest Lecturer, Milwaukee, WI, December 2023.

Formal Bayesian Technique to Measure Unobserved FMRI Data for Quicker Brain Imaging. Office of Research and Sponsored Programs, Marquette University, Forward Thinking Symposium, November 2022.

Increasing the Accuracy of Statistical Analysis and Task Activation with a Full Bayesian Approach to SENSE Image Reconstruction. Department of Mathematical and Statistical Sciences, Marquette University, Department Colloquium, Milwaukee, WI, September 2022.

Bayesian Complex Valued Latent Regression. Department of Mathematical and Statistical Sciences, Marquette University, Guest Lecturer, Milwaukee, WI, November 2021.

A Formal Bayesian Approach to SENSE Image Reconstruction. Department of Mathematical and Statistical Sciences, Marquette University, Department Colloquium, Milwaukee, WI, October 2021.

Posters

Functional data analysis of spatial protein imaging data using spatial trajectories with application to ovarian cancer. Research Month, Children's Mercy, Kansas City MO, May 2026.

Bayesian Mixture Model for Immune Subtyping Using Immune Counts from Spatial Proteomic Imaging Data. Frontiers Research Symposium, Kansas City MO, March 2026.

Assessment of Different Relationship Structures Between Immune Cell Populations within a Bayesian Hierarchical Model for Spatial Protein Imaging Data. KUCC Research Symposium, Overland Park, KS, December 2025.

Bayesian Modeling of Immune Cell Populations Within the Tumor Microenvironment with Application to Ovarian Cancer. KUCC Cancer Biology Research Meeting, Olathe, KS, July 2025.

Bayesian Hierarchical Modeling that Leverages the Relationships Between Immune Cell Populations Measured from Spatial Proteomic Imaging Technologies. Research Month, Children's Mercy, Kansas City, MO, 2025.

TEACHING PHILOSOPHY

My **mission as an educator** is to help students develop confidence as quantitative thinkers by making statistical reasoning rigorous, accessible, and relevant. My **teaching approach** emphasizes conceptual understanding, active problem solving, and authentic applications that connect statistical theory to meaningful scientific questions. Through clear expectations, individualized support, and reproducible computational practice, I strive to create an inclusive learning environment in which students develop both technical skills and intellectual independence. My **vision** is for students to leave my courses prepared to think critically about data, communicate uncertainty responsibly, and apply statistical reasoning thoughtfully in research, professional practice, and everyday decision-making.

TEACHING EXPERIENCE

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| 11/2026 | <p>Introduction to R/RStudio, Children's Mercy Kansas City, Population Health & Outcomes Research, Fridley Lab, Kansas City, MO</p> <ul style="list-style-type: none">• Co-developed and co-instructed an institute-wide introductory R/RStudio course for physicians, trainees, and research staff, providing hands-on instruction in statistical programming, reproducible research, and data analysis workflows. |
| 08/2023 – 05/2024 | <p>Student Success Coordinator – Modern Elementary Statistics, Marquette University, Department of Mathematical and Statistical Sciences, Milwaukee, WI</p> <ul style="list-style-type: none">• Led weekly review sessions supporting student success across multiple sections of Modern Elementary Statistics.• Provided individualized academic support through one-on-one mentoring and tutoring to improve student performance and engagement.• Developed practice materials, homework assignments, and instructional resources to promote consistency across course sections. |
| 08/2020 – 05/2021;
08/2022 – 05/2023 | <p>Graduate Teaching Assistant, Marquette University, Department of Mathematical and Statistical Sciences, Milwaukee, WI</p> <ul style="list-style-type: none">• Facilitated discussion sessions, office hours, and student support activities for Modern Elementary Statistics, Honors Modern Elementary Statistics, and Statistical Methods.• Assisted with grading, assessment, and course administration for undergraduate statistics courses.• Provided individualized instruction and mentoring to strengthen students' understanding of statistical reasoning, data analysis, and quantitative problem solving. |

PROFESSIONAL DEVELOPMENT

TL1 Training Activities

- 2025 – Present **Coursework Completed/Planned:** Grant Writing (Fall 2025), Responsible Conduct of Research (Spring 2026), Principles of Epidemiology (Spring 2027), Scientific Writing (Spring 2027)
- 2025 – Present **Monthly Scholars Club Meetings:** Research topic discussions, career development presentations/activities, and Clifton Strengths Leadership Training

Workshops

- 2025 & 2026 **Midwest Research Skills Development Workshop**, University of Kansas Medical Center. Topics included mentoring, grantsmanship, negotiation, time management, work-life balance, and professional development

Short Courses

- 09/2026 **Introduction to Multi-modal Pathology Foundation Models**, American Statistical Association, Section on Statistics in Genomics and Genetics, Webinar Series
- 08/2026 **Statistics and Machine Learning for Geospatial Analysis**, American Statistical Association, Joint Statistical Meeting
- 08/2026 **Introduction to Python for R Users**, American Statistical Association, Joint Statistical Meeting
- 06/2025 **Genomic Medicine Short Course**, Children's Mercy Kansas City, Genomic Medicine Center

SERVICE PHILOSOPHY

My **mission in service** is to strengthen the academic and scientific communities that support rigorous, collaborative, and reproducible research. My **service agenda** includes thoughtful peer review, student and colleague mentorship, professional engagement, interdisciplinary collaboration, and the development of open-source tools and resources that make statistical methods more accessible and impactful. I view service as an essential component of academic citizenship and a means of advancing the quality, accessibility, and integrity of scientific discovery. My **vision** is a scholarly community in which researchers contribute not only through their own achievements, but also by fostering environments that enable colleagues, students, institutions, and the broader field to thrive.

SERVICE

Non-Institutional Committee Memberships

- 2026 – Present Member, ASA Section on Statistics in Genetics and Genomics (SSGG)
Member Engagement Committee

Journal Reviewer

IEEE Transactions on Medical Imaging
Journal of Molecular Medicine
PLOS Computational Biology

Translational Oncology

Professional Memberships

American Association for Cancer Research
Eastern North American Region International Biometric Society
American Statistical Association
International Society for Bayesian Analysis
Honor Society of Phi Kappa Phi
Kappa Mu Epsilon Mathematical Honor Society



*My **mission** is to develop rigorous statistical methods, educate the next generation of quantitative scientists, and foster collaborative scientific communities that advance reproducible research and precision medicine.*

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